Fixed-shift Liouville pair correlations: exact reductions, projected transport, and local quadratic obstructions

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For a fixed nonzero shift h, we study the Liouville pair-correlation sums

$$A_h(N) := \sum_{n \le N} \lambda(n)\lambda(n+h).$$

We develop a critical-scale reduction theory for the asymptotic formula $A_h(N) = o(N)$, the fixed-shift two-point case of Chowla's conjecture, at the scale $H = X^{1/2}$. Passing to the derived sequence $a_h(n) = \lambda(n)\lambda(n+h)$, we separate the direct absolute first-moment implication from the signed Fejér-kernel and dyadic-shell identities, and we record exact quartic Cesàro, Fourier–Dirichlet, dyadic-shell, and first-moment reformulations of the critical short-block criterion. We then prove the correct fixed-shift local U^2 reduction: the normalized local quantity is a fourth-power quantity, so block sums are controlled through a fourth-power estimate and a Young/Hölder averaging step, not by a false linear pointwise bound. Finally we introduce a projected packet formalism for bad blocks. On the projected side we establish an exact affine-shear identity, a rigidity theorem, and an exact transport-defect identity, and we show that a large projected Fejér/shear packet forces a local quadratic-phase obstruction for λ. The output is a structured conditional reduction of the remaining obstruction to projected bias, transport rigidity, and projected long-range packet formation.

1 Introduction and statement of results

Let $\lambda(n) = (-1)^{\Omega(n)}$ be the Liouville function. For a fixed nonzero shift $h \in \mathbb{Z}$, we consider the pair-correlation sum

$$A_h(N) := \sum_{n \le N} \lambda(n)\lambda(n+h).$$

The asymptotic formula

$$A_h(N) = o(N) \qquad (N \to \infty)$$

is the fixed-shift two-point case of Chowla's conjecture [6]. Major progress on correlations of multiplicative functions is now available in logarithmically averaged settings, at almost all scales, and through short-interval, Fourier-uniformity, and higher-uniformity theorems

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on average [1, 2, 3, 4, 5, 11, 12]. The fixed-shift pointwise problem nevertheless remains especially rigid. The purpose of this paper is to isolate that rigidity at the critical scale

$$H = X^{1/2},$$

and to show that the remaining obstruction can be expressed in a sharply structured local form.

Our analysis is carried out entirely at one fixed shift. Starting from the critical short-block mean-square criterion of Section 2, we pass to the derived sequence

$$a_h(n) := \lambda(n)\lambda(n+h),$$

and develop a chain of exact reformulations and reductions for its short sums and associated energy profile. In particular, we obtain quartic Cesàro, Fourier–Dirichlet, dyadic-shell, and short-interval first-moment criteria, together with a positive multiscale absorption theorem. The Fourier-kernel side of the argument is classical in spirit [7, 10], while the multiscale organization is deterministic and chaining-style rather than generic chaining in Talagrand's stochastic-process sense [13]. We then record a fixed-shift local U^2 reduction in the fourth-power normalization used in the formal development. This point is important: the averaged local U^2 input is consumed through a fourth-power estimate and a scalar Young inequality, not through a linear pointwise bound in the normalized fourth-power mass. Finally we introduce a projected packet formalism for bad blocks. On the projected side we establish an exact affine-shear identity, a rigidity theorem, and an exact transport-defect identity, and we show that a large projected Fejér/shear packet forces a local quadratic-phase obstruction for λ, in the broader context of recent polynomial-phase and higher-uniformity results for multiplicative functions in short intervals [5, 11, 12]. In this way the fixed-shift problem is reduced to a precise final dichotomy between projected bias and projected long-range packet formation.

A basic global consequence of the short-block framework is the following conditional implication. It is stated in absolute first-moment form; this is stronger information than signed Fejér cancellation, because it controls the total mass of bad short blocks rather than cancellations between shells.

Theorem 1.1 (Conditional fixed-shift theorem)**.** *Fix $h \neq 0$. Assume that, with $H = \lfloor X^{1/2} \rfloor$, one has*

$$\sup_{1 \le M \le H} \frac{1}{XM} \sum_{X \le x \le 2X-M} \left| \sum_{x<n\le x+M} \lambda(n)\lambda(n+h) \right| \to 0 \qquad (X \to \infty). \tag{1.1}$$

Then

$$A_h(N) = o(N) \qquad (N \to \infty).$$

Section 6 contains the direct first-moment implication and its dyadic-shell consequence. Section 7 records the corrected fourth-power local U^2 reduction. Section 8 develops the projected transport and packet framework that captures the final transition in the reduction.

2 The critical short-block criterion

Fix $h \neq 0$ and define

$$a_h(n) := \lambda(n)\lambda(n+h), \qquad A_h(N) := \sum_{n \le N} a_h(n),$$

and for integers x and $H \geq 1$ define

$$B_h(x;H) := \sum_{x<n\leq x+H} a_h(n) = A_h(x+H) - A_h(x).$$

Theorem 2.1 (Critical short-block criterion)**.** *Fix $h \neq 0$. Assume that for $H = \lfloor X^{1/2} \rfloor$ one has*

$$\sum_{X\leq x\leq 2X-H} |B_h(x;H)|^2 = o(XH^2) \qquad (X \to \infty). \tag{2.1}$$

Then

$$A_h(N) = o(N) \qquad (N \to \infty). \tag{2.2}$$

Proof. Write

$$\eta(X) := \frac{1}{XH^2} \sum_{X\leq x\leq 2X-H} |B_h(x;H)|^2,$$

so that $\eta(X) \to 0$ by (2.1). Fix a parameter $0 < \tau \leq 1$, to be chosen later, and define the bad set

$$\mathcal{E}_\tau := \{x \in [X, 2X-H] \cap \mathbb{Z} : \ |B_h(x;H)| > \tau H\}.$$

By Markov's inequality,

$$\#\mathcal{E}_\tau \cdot \tau^2 H^2 \leq \sum_{X\leq x\leq 2X-H} |B_h(x;H)|^2 = \eta(X)XH^2,$$

so

$$\#\mathcal{E}_\tau \leq \eta(X)\tau^{-2}X. \tag{2.3}$$

Now split starting points by residue class modulo H. For $r \in \{0, 1, \ldots, H-1\}$ set

$$x_{r,j} := X + r + jH.$$

Averaging over r shows that there exists a residue class $r = r_X$ such that

$$\#\{j : \ x_{r,j} \in \mathcal{E}_\tau\} \leq \frac{\#\mathcal{E}_\tau}{H} \leq \eta(X)\tau^{-2}\frac{X}{H}.$$

Fix such an r and write $x_j := x_{r,j}$.

Let $N \in [X, 2X]$, and choose m so that

$$x_m \leq N < x_{m+1},$$

with the convention $m = -1$ if $N < x_0$. Then

$$A_h(N) - A_h(x_0) = \sum_{j=0}^{m-1} (A_h(x_{j+1}) - A_h(x_j)) + O(H) = \sum_{j=0}^{m-1} B_h(x_j;H) + O(H),$$

since the tail interval between x_m and N has length $< H$ and $|a_h(n)| \leq 1$.

Among the blocks x_j, every good block contributes at most τH in absolute value, while the total number of bad blocks is at most $\eta(X)\tau^{-2}X/H$. Since $mH \ll X$, we obtain

$$|A_h(N) - A_h(x_0)| \ll \tau X + \eta(X)\tau^{-2}X + H.$$

Also $x_0 = X + r$ with $0 \leq r < H$, so

$$|A_h(x_0) - A_h(X)| \leq H.$$

Hence

$$\sup_{X \le N \le 2X} |A_h(N) - A_h(X)| \ll \tau X + \eta(X)\tau^{-2}X + H. \tag{2.4}$$

Choose $\tau := \eta(X)^{1/3}$. Then

$$\tau X + \eta(X)\tau^{-2}X \ll \eta(X)^{1/3}X,$$

and because $H = X^{1/2} + O(1) = o(X)$, (2.4) becomes

$$\sup_{X \le N \le 2X} |A_h(N) - A_h(X)| = o(X). \tag{2.5}$$

Now let $X_j := 2^j$. For any N with $2^K \le N < 2^{K+1}$,

$$A_h(N) = A_h(1) + \sum_{j=0}^{K-1} \left(A_h(2^{j+1}) - A_h(2^j)\right) + \left(A_h(N) - A_h(2^K)\right).$$

Applying (2.5) on each dyadic interval $[2^j, 2^{j+1}]$ shows that

$$A_h(2^{j+1}) - A_h(2^j) = o(2^j), \qquad A_h(N) - A_h(2^K) = o(2^K).$$

Therefore, for every $\varepsilon > 0$, all sufficiently large j satisfy

$$\left|A_h(2^{j+1}) - A_h(2^j)\right| \le \varepsilon 2^j, \qquad \left|A_h(N) - A_h(2^K)\right| \le \varepsilon 2^K,$$

so

$$|A_h(N)| \le O_\varepsilon(1) + \varepsilon \sum_{j=0}^{K} 2^j \ll O_\varepsilon(1) + \varepsilon N.$$

Dividing by N and letting $N \to \infty$, then $\varepsilon \to 0$, gives (2.2). □

3 Energy profile and quartic correlations

We now pass from the short-block criterion to the exact quadratic and quartic identities attached to the derived sequence a_h. Throughout this section, let $X \ge 1$ be large and set

$$H := \lfloor X^{1/2} \rfloor.$$

For integers $M \ge 1$, define the short-block energy

$$E_h(M; X) := \sum_{X \le x \le 2X - M} |B_h(x; M)|^2.$$

Also define the quartic correlation sums

$$P_h(t; X) := \sum_{X < n \le 2X - t} a_h(n)a_h(n+t) \qquad (t \ge 1),$$

and the Cesàro partial sums

$$C_h(M; X) := \sum_{1 \le t \le M} P_h(t; X).$$

Proposition 3.1 (Exact energy profile)**.** *For every integer $M \ge 1$,*

$$E_h(M; X) = MX + 2\sum_{1 \le t < M} (M - t)P_h(t; X) + O_h(M^3). \tag{3.1}$$

Proof. Let

$$a_{h,X}(n) := a_h(n)\,\mathbf{1}_{X<n\le 2X}.$$

Set

$$\widetilde{B}_h(x;M) := \sum_{x<n\le x+M} a_{h,X}(n).$$

For $X \le x \le 2X - M$ one has $\widetilde{B}_h(x;M) = B_h(x;M)$. Outside this range there are only $O(M)$ values of x for which $\widetilde{B}_h(x;M) \ne 0$, and trivially $\left|\widetilde{B}_h(x;M)\right| \le M$. Hence

$$\sum_{x\in\mathbb{Z}} \left|\widetilde{B}_h(x;M)\right|^2 = E_h(M;X) + O(M^3). \tag{3.2}$$

Expanding the square gives the exact block-energy identity

$$\sum_{x\in\mathbb{Z}} \left|\widetilde{B}_h(x;M)\right|^2 = \sum_{|t|<M} (M-|t|) \sum_{n\in\mathbb{Z}} a_{h,X}(n)a_{h,X}(n+t).$$

Since $X > |h|$ for X large, we have $a_h(n)^2 = 1$ on $(X, 2X]$, and so

$$\sum_{n\in\mathbb{Z}} a_{h,X}(n)^2 = X + O_h(1).$$

For $t \ge 1$,

$$\sum_{n\in\mathbb{Z}} a_{h,X}(n)a_{h,X}(n+t) = P_h(t;X),$$

and the contribution of negative t is the same by symmetry. Therefore

$$\sum_{x\in\mathbb{Z}} \left|\widetilde{B}_h(x;M)\right|^2 = MX + 2\sum_{1\le t<M} (M-t)P_h(t;X) + O_h(M).$$

Combining this with (3.2) proves (3.1). □

Corollary 3.2 (Discrete derivative of the energy profile)**.** *For every integer* $1 \le M \le H$*,*

$$E_h(M+1;X) - E_h(M;X) = X + 2C_h(M;X) + O_h(M^2). \tag{3.3}$$

Proof. Subtract (3.1) at $M+1$ and M. The diagonal term contributes X, while the quartic terms telescope to $2\sum_{1\le t\le M} P_h(t;X) = 2C_h(M;X)$. The error term changes by $O_h(M^2)$. □

Corollary 3.3 (A uniform slope criterion)**.** *If*

$$\sup_{1\le M\le H} |E_h(M+1;X) - E_h(M;X) - X| = o(XH),$$

then (2.2) *holds.*

Proof. By (3.3),

$$\sup_{1\le M\le H} |C_h(M;X)| = o(XH)$$

(because $M^2 \ll H^2 \asymp X = o(XH)$). Then

$$\sum_{1\le t<H} (H-t)P_h(t;X) = \sum_{M=1}^{H-1} C_h(M;X) = o(XH^2),$$

and now (3.1) at $M = H$ yields

$$E_h(H;X) = HX + o(XH^2) + O(H^3) = o(XH^2),$$

since $HX \asymp X^{3/2} = o(X^2) = o(XH^2)$ and $H^3 \asymp X^{3/2} = o(XH^2)$. The conclusion follows from Theorem 2.1. □

Remark 3.4 (Signed Fejér cancellation and Abel summation)**.** The preceding identities also explain a useful boundary of the method. If

$$S_h(M;X) := \sum_{1\le t\le M} P_h(t;X),$$

then the Fejér-weighted off-diagonal term satisfies the exact Abel identity

$$\sum_{1\le t<H} (H-t)P_h(t;X) = \sum_{M=1}^{H-1} S_h(M;X).$$

Consequently a signed hypothesis such as

$$\sup_{1\le M\le H} |S_h(M;X)| = o(XH)$$

would imply the critical energy estimate by a direct summation-by-parts argument, together with Proposition 3.1. This signed route is deliberately not the same as the absolute first-moment hypothesis used in Theorem 1.1: signed Fejér cancellation can hide cancellation between positive and negative shell contributions, whereas the absolute first moment controls the total mass of large short blocks and feeds into the bad-block and packet reductions below.

4 Fourier reformulation and positive multiscale absorption

The discrete derivative identity naturally leads to a Fourier reformulation. The short-block energies are represented by Fejér kernels, while their discrete derivatives are represented by Dirichlet kernels; this is standard Fourier analysis on the torus [7, 10]. This is the point at which the oscillatory obstruction is converted into a positive multiscale quantity.

Define

$$A_{h,X}(\alpha) := \sum_{X<n\le 2X} a_h(n)e(\alpha n)\,, \qquad \alpha\in\mathbb{R}/\mathbb{Z}.$$

For $M\ge 1$, let

$$F_M(\alpha) := \left|\sum_{m=1}^{M} e(\alpha m)\right|^2, \qquad D_M(\alpha) := \sum_{|t|\le M} e(\alpha t)\,.$$

Proposition 4.1 (Fejér representation)**.** *For every integer $M\ge 1$,*

$$E_h(M;X) = \int_0^1 |A_{h,X}(\alpha)|^2\, F_M(\alpha)\, d\alpha + O_h(M^3). \tag{4.1}$$

Moreover, for $1\le M\le H$,

$$E_h(M+1;X) - E_h(M;X) = \int_0^1 |A_{h,X}(\alpha)|^2\, D_M(\alpha)\, d\alpha + O_h(M^2). \tag{4.2}$$

Proof. Let $K_M(n) := \mathbf{1}_{1\le n\le M}$. Then

$$\widetilde{B}_h(x;M) = (a_{h,X} * K_M)(x),$$

where $a_{h,X}$ is as in the proof of Proposition 3.1. By Parseval on $\mathbb{Z}$,

$$\sum_{x\in\mathbb{Z}} \left|\widetilde{B}_h(x;M)\right|^2 = \int_0^1 |A_{h,X}(\alpha)|^2 \left|\widehat{K}_M(\alpha)\right|^2 d\alpha = \int_0^1 |A_{h,X}(\alpha)|^2\, F_M(\alpha)\, d\alpha.$$

Using (3.2) proves (4.1). The identity

$$F_{M+1}(\alpha) - F_M(\alpha) = D_M(\alpha)$$

follows by comparing Fourier coefficients, and then (4.2) follows by subtraction. □

Lemma 4.2 (A positive multiscale domination). *For every integer $H \geq 1$ and every $\alpha \in \mathbb{R}/\mathbb{Z}$,*

$$\sup_{1 \leq L \leq H} |D_L(\alpha) - 1| \ll \sum_{M=1}^{H} \frac{F_M(\alpha)}{M^2}. \tag{4.3}$$

Proof. Let $\delta := \|\alpha\|$, where $\|\alpha\|$ denotes the distance from α to the nearest integer. The standard Dirichlet-kernel bound gives

$$\sup_{1 \leq L \leq H} |D_L(\alpha) - 1| \ll \min(H, \delta^{-1}).$$

It therefore suffices to show

$$\sum_{M=1}^{H} \frac{F_M(\alpha)}{M^2} \gg \min(H, \delta^{-1}).$$

If $\delta \leq 1/(2H)$, then $\delta \leq 1/(2M)$ for every $1 \leq M \leq H$, and the usual estimate

$$F_M(\alpha) = \left| \frac{\sin(\pi M \alpha)}{\sin(\pi \alpha)} \right|^2 \gg M^2$$

shows that the sum is $\gg H$.

If $\delta > 1/(2H)$, let $M_0 := \lfloor (2\delta)^{-1} \rfloor$. Then $1 \leq M_0 \leq H$ and still $\delta \leq 1/(2M)$ for every $1 \leq M \leq M_0$, so $F_M(\alpha) \gg M^2$ throughout that range. Hence

$$\sum_{M=1}^{H} \frac{F_M(\alpha)}{M^2} \gg M_0 \asymp \delta^{-1}.$$

This proves (4.3). □

Theorem 4.3 (Positive multiscale criterion). *Assume that*

$$\sum_{M=1}^{H} \frac{E_h(M;X)}{M^2} = o(XH) \qquad (X \to \infty). \tag{4.4}$$

Then (2.2) *holds.*

Proof. By Proposition 4.1,

$$\int_0^1 |A_{h,X}(\alpha)|^2 \, d\alpha = \sum_{X < n \leq 2X} a_h(n)^2 = X + O_h(1).$$

Hence, by Corollary 3.3, it is enough to show that

$$\sup_{1 \leq L \leq H} \left| \int_0^1 |A_{h,X}(\alpha)|^2 \, (D_L(\alpha) - 1) \, d\alpha \right| = o(XH).$$

Applying Lemma 4.2, then interchanging sum and integral, and finally using Proposition 4.1, we obtain

$$\sup_{1 \leq L \leq H} \left| \int_0^1 |A_{h,X}(\alpha)|^2 \, (D_L(\alpha) - 1) \, d\alpha \right| \ll \sum_{M=1}^{H} \frac{1}{M^2} \int_0^1 |A_{h,X}(\alpha)|^2 \, F_M(\alpha) \, d\alpha$$

$$= \sum_{M=1}^{H} \frac{E_h(M;X)}{M^2} + O\left(\sum_{M=1}^{H} M \right) = \sum_{M=1}^{H} \frac{E_h(M;X)}{M^2} + O(H^2).$$

Since $H^2 \asymp X = o(XH)$, hypothesis (4.4) yields the desired bound. □

5 Dyadic shells, tree weights, and harmonic weights

We next reorganize the multiscale sum shell-by-shell in the shift variable t. This isolates the precise dyadic local cancellation required of the quartic family.

Let $J \geq 0$ be defined by

$$2^{J-1} < H \leq 2^J.$$

For $0 \leq k \leq J-1$, write

$$I_k := \{t \in \mathbb{Z} : 2^k \leq t < 2^{k+1}\}.$$

For integers $u \in I_k$, define the shell partial sums

$$Q_{h,k}(u; X) := \sum_{2^k \leq t \leq u} P_h(t; X).$$

Also define the tree-stopped multiscale energy

$$\mathcal{T}_h(X) := \sum_{j=0}^{J} \frac{E_h(2^j; X)}{2^j}.$$

Proposition 5.1 (Tree weight identity)**.** *One has*

$$\mathcal{T}_h(X) = X(J+1) + 2 \sum_{1 \leq t < 2^J} \omega_J(t) P_h(t; X) + O_h(H^2), \tag{5.1}$$

where

$$\omega_J(t) := \sum_{\substack{0 \leq j \leq J \\ 2^j > t}} \left(1 - \frac{t}{2^j}\right).$$

If $t \in I_k$, then

$$\omega_J(t) = A_k - B_k t, \qquad A_k := J - k, \qquad B_k := \sum_{j=k+1}^{J} 2^{-j}. \tag{5.2}$$

In particular,

$$A_k \ll J - k + 1, \qquad B_k \asymp 2^{-k}. \tag{5.3}$$

Proof. Insert (3.1) with $M = 2^j$ into the definition of $\mathcal{T}_h(X)$ and interchange the order of summation. For $t \in I_k$, the condition $2^j > t$ is equivalent to $j \geq k+1$, which gives the affine formula (5.2). The size bounds are immediate. □

Proposition 5.2 (Harmonic weight identity)**.** *Define*

$$\Sigma_h(X) := \sum_{M=1}^{H} \frac{E_h(M; X)}{M^2}.$$

Then

$$\Sigma_h(X) = X \sum_{M=1}^{H} \frac{1}{M} + 2 \sum_{1 \leq t < H} \nu_H(t) P_h(t; X) + O_h(H^2), \tag{5.4}$$

where

$$\nu_H(t) := \sum_{M=t+1}^{H} \frac{M-t}{M^2} = \sum_{M=t+1}^{H} \frac{1}{M} - t \sum_{M=t+1}^{H} \frac{1}{M^2}.$$

Moreover, if $t \in I_k$, then

$$\nu_H(t) \ll J - k + 1, \tag{5.5}$$

and for $2^k \le u < 2^{k+1} - 1$*,*

$$\nu_H(u) - \nu_H(u+1) = \sum_{M=u+1}^{H} \frac{1}{M^2} \asymp 2^{-k}. \tag{5.6}$$

Proof. Insert (3.1) into the definition of $\Sigma_h(X)$ and interchange the order of summation:

$$\Sigma_h(X) = X \sum_{M=1}^{H} \frac{1}{M} + 2 \sum_{1 \le t < H} \left(\sum_{M=t+1}^{H} \frac{M-t}{M^2} \right) P_h(t;X) + O_h\left(\sum_{M=1}^{H} M \right).$$

Since $\sum_{M=1}^{H} M \ll H^2$, this is (5.4).

If $t \in I_k$, then

$$\nu_H(t) \le \sum_{M=t+1}^{H} \frac{1}{M} \ll \log \frac{H}{t+1} + 1 \ll J - k + 1,$$

which gives (5.5). Finally,

$$\nu_H(u) - \nu_H(u+1) = \frac{1}{(u+1)^2} + \sum_{M=u+2}^{H} \frac{1}{M^2} = \sum_{M=u+1}^{H} \frac{1}{M^2},$$

and since $u \asymp 2^k$ on I_k, this is $\asymp 2^{-k}$. This proves (5.6). □

Theorem 5.3 (Dyadic local quartic criterion)**.** *Assume that*

$$\sup_{\substack{0 \le k \le J-1 \\ u \in I_k}} \frac{|Q_{h,k}(u;X)|}{X 2^k} \to 0 \qquad (X \to \infty). \tag{5.7}$$

Then (4.4) *holds, hence* (2.2) *holds.*

Proof. We apply shellwise summation by parts to the weighted quartic sum in (5.4). For a fixed shell $I_k = [2^k, 2^{k+1} - 1]$, write

$$\Sigma_k := \sum_{t \in I_k} \nu_H(t) P_h(t;X).$$

Then

$$\Sigma_k = \nu_H(2^{k+1} - 1) Q_{h,k}(2^{k+1} - 1; X) + \sum_{u=2^k}^{2^{k+1}-2} (\nu_H(u) - \nu_H(u+1)) Q_{h,k}(u;X).$$

Using (5.5), (5.6), and hypothesis (5.7), we obtain

$$\begin{aligned} |\Sigma_k| &\ll (J-k+1) \sup_{u \in I_k} |Q_{h,k}(u;X)| + 2^{-k} \sum_{u=2^k}^{2^{k+1}-2} \sup_{v \in I_k} |Q_{h,k}(v;X)| \\ &\ll (J-k+1) \sup_{u \in I_k} |Q_{h,k}(u;X)| = o\big(X(J-k+1)2^k\big). \end{aligned}$$

Summing over k and using

$$\sum_{k=0}^{J-1} (J-k+1) 2^k \ll 2^J \asymp H,$$

we deduce

$$\sum_{1\le t<H} \nu_H(t)P_h(t;X) = o(XH).$$

Also

$$X\sum_{M=1}^{H}\frac{1}{M} \ll X\log H = o(XH), \qquad H^2 = o(XH).$$

Hence (5.4) gives

$$\Sigma_h(X) = o(XH),$$

which is exactly (4.4). The conclusion now follows from Theorem 4.3. □

Remark 5.4. The dyadic local criterion isolates a scale-by-scale version of the quartic cancellation problem. The input is local in the t-variable and is therefore naturally compatible with shellwise stopping and summation-by-parts arguments.

6 A short-interval first-moment theorem

We now isolate the role of the absolute first moment. There is a direct implication from

$$\sup_{1\le M\le H}\frac{1}{XM}\sum_{X\le x\le 2X-M}|B_h(x;M)| = o(1)$$

to the critical short-block criterion: at the single scale $M = H$, the pointwise bound $|B_h(x;H)| \le H$ gives

$$\sum_{X\le x\le 2X-H}|B_h(x;H)|^2 \le H\sum_{X\le x\le 2X-H}|B_h(x;H)| = o(XH^2).$$

Thus the first-moment implication itself is elementary once absolute values are available. We nevertheless keep the dyadic-shell bridge because it identifies exactly which local quartic shell quantities are controlled by the same absolute first moments, and these shell quantities feed the packet analysis below.

Proposition 6.1 (Shells are dominated by first moments)**.** *Fix $0 \le k \le J-1$ and $u \in I_k$. Set*

$$T := u - 2^k + 1 \qquad (1 \le T \le 2^k).$$

Then

$$|Q_{h,k}(u;X)| \le \sum_{X\le x\le 2X-T}|B_h(x;T)| + O_h(T^2). \tag{6.1}$$

Proof. Expand the definition of $Q_{h,k}(u;X)$:

$$Q_{h,k}(u;X) = \sum_{t=2^k}^{u}\sum_{X<n\le 2X-t} a_h(n)a_h(n+t).$$

Interchanging the order of summation gives

$$Q_{h,k}(u;X) = \sum_{X<n\le 2X-2^k} a_h(n)\sum_{2^k\le t\le \min(u,2X-n)} a_h(n+t).$$

Split at $n \le 2X-u$. The main range contributes

$$\sum_{X<n\le 2X-u} a_h(n)\sum_{t=2^k}^{u} a_h(n+t),$$

and the inner sum is exactly $B_h(n+2^k-1;T)$. Taking absolute values yields

$$\left| \sum_{X<n\le 2X-u} a_h(n) \sum_{t=2^k}^{u} a_h(n+t) \right| \le \sum_{X\le x\le 2X-T} |B_h(x;T)|.$$

The boundary range $2X-u<n\le 2X-2^k$ has at most $T-1$ values of n, and for each such n the inner sum has length at most T, so its total contribution is $O(T^2)$. This proves (6.1). □

Theorem 6.2 (Short-interval first-moment theorem)**.** *Assume that*

$$\sup_{1\le M\le H} \frac{1}{XM} \sum_{X\le x\le 2X-M} |B_h(x;M)| \to 0 \qquad (X\to\infty). \tag{6.2}$$

Then (2.2) *holds.*

Proof. Apply (6.2) at $M=H$. Since $|a_h(n)|\le 1$, every H-block satisfies $|B_h(x;H)|\le H$. Hence

$$\sum_{X\le x\le 2X-H} |B_h(x;H)|^2 \le H \sum_{X\le x\le 2X-H} |B_h(x;H)| = o(XH^2).$$

The critical short-block criterion, Theorem 2.1, now gives (2.2). □

Corollary 6.3 (First moments also control dyadic local quartic shells)**.** *Under the hypothesis* (6.2)*, the dyadic local quartic condition* (5.7) *holds.*

Proof. Fix a shell I_k and a point $u\in I_k$, and let

$$T := u-2^k+1 \qquad (1\le T\le 2^k\le H).$$

By Proposition 6.1,

$$|Q_{h,k}(u;X)| \le \sum_{X\le x\le 2X-T} |B_h(x;T)| + O_h(T^2).$$

The hypothesis (6.2) applies uniformly for every $1\le T\le H$, so

$$\sum_{X\le x\le 2X-T} |B_h(x;T)| = o(XT).$$

Since $T\le H\asymp X^{1/2}$, the boundary term also satisfies

$$T^2\le HT = o(XT).$$

Hence

$$|Q_{h,k}(u;X)| = o(XT) = o(X2^k)$$

uniformly in k and u, which is exactly (5.7). □

7 A fixed-shift local U^2 reduction

We next record the local U^2 reduction in the normalization used in the formal development. The point which matters for the sequel is that the normalized local quantity below is a fourth-power quantity. Consequently the correct block estimate is a fourth-power estimate, not a linear estimate in the normalized fourth-power mass.

For a finitely supported function $f : \mathbb{Z} \to \mathbb{C}$ and a finite interval $I \subset \mathbb{Z}$, put

$$C_f(I;t) := \sum_{n\in I} \mathbf{1}_I(n+t) f(n+t)\overline{f(n)}.$$

Define

$$U_2^4(f;I) := \sum_{-|I|\le t\le |I|} |C_f(I;t)|^2, \qquad \mathcal{U}_2(f;I) := \frac{U_2^4(f;I)}{\max(|I|,1)^3}.$$

For $x, M \in \mathbb{N}$, let

$$I_{x,M} := \{x+1, x+2, \dots, x+M\} \subset \mathbb{Z}$$

and let $\widetilde{a} : \mathbb{Z} \to \mathbb{C}$ be the zero extension of a from $\mathbb{N}$ to $\mathbb{Z}$, viewed as complex-valued. Thus

$$B_a(x;M) = \sum_{x<n\le x+M} a(n) = \sum_{n\in I_{x,M}} \widetilde{a}(n).$$

When $a = a_h$, we write $B_h(x;M)$ as before.

Proposition 7.1 (Correlation expansion)**.** *For every $a : \mathbb{N} \to \mathbb{R}$ and every $x, M \in \mathbb{N}$,*

$$\sum_{-M\le t\le M} C_{\widetilde{a}}(I_{x,M};t) = \left(\sum_{n\in I_{x,M}} \widetilde{a}(n)\right)\overline{\left(\sum_{n\in I_{x,M}} \widetilde{a}(n)\right)}.$$

Equivalently,

$$\sum_{-M\le t\le M} C_{\widetilde{a}}(I_{x,M};t) = B_a(x;M)\overline{B_a(x;M)}.$$

Proof. Expand the left-hand side and interchange the two finite sums:

$$\sum_{-M\le t\le M}\ \sum_{n\in I_{x,M}} \mathbf{1}_{I_{x,M}}(n+t)\widetilde{a}(n+t)\overline{\widetilde{a}(n)}.$$

For fixed $n \in I_{x,M}$, the change of variables $m = n+t$ bijects the admissible shifts with $m \in I_{x,M}$. Hence the inner sum is

$$\sum_{m\in I_{x,M}} \widetilde{a}(m)\overline{\widetilde{a}(n)}.$$

Summing in n gives

$$\sum_{n\in I_{x,M}} \left(\sum_{m\in I_{x,M}} \widetilde{a}(m)\right)\overline{\widetilde{a}(n)} = \left(\sum_{m\in I_{x,M}} \widetilde{a}(m)\right)\overline{\left(\sum_{n\in I_{x,M}} \widetilde{a}(n)\right)}.$$

$\square$

Proposition 7.2 (Fourth-power local U^2 block estimate)**.** *For every $a : \mathbb{N} \to \mathbb{R}$, every $x \in \mathbb{N}$, and every $M \ge 1$,*

$$|B_a(x;M)|^4 \le 3M^4\,\mathcal{U}_2(\widetilde{a}; I_{x,M}).$$

Proof. By Proposition 7.1,

$$|B_a(x;M)|^4 = \left| \sum_{-M\le t\le M} C_{\widetilde{a}}(I_{x,M};t) \right|^2 .$$

Cauchy's inequality gives

$$\left| \sum_{-M\le t\le M} C_{\widetilde{a}}(I_{x,M};t) \right|^2 \le (2M+1) \sum_{-M\le t\le M} \left|C_{\widetilde{a}}(I_{x,M};t)\right|^2 .$$

Since $M \ge 1$, $2M+1 \le 3M$. Therefore

$$|B_a(x;M)|^4 \le 3M\, U_2^4(\tilde{a}; I_{x,M}).$$

Finally $|I_{x,M}| = M$, so

$$U_2^4(\tilde{a}; I_{x,M}) = M^3 \mathcal{U}_2(\tilde{a}; I_{x,M}),$$

which gives the claimed bound. □

Remark 7.3 (The missing fourth root)**.** Because $\mathcal{U}_2$ is normalized as a fourth-power quantity, Proposition 7.2 implies only

$$|B_a(x;M)| \le 3^{1/4} M\, \mathcal{U}_2(\tilde{a}; I_{x,M})^{1/4}.$$

A pointwise estimate of the form

$$|B_a(x;M)| \le CM\, \mathcal{U}_2(\tilde{a}; I_{x,M})$$

would use the fourth-power normalized quantity linearly. That is not the estimate used below. The averaged first-moment implication is instead obtained from the fourth-power estimate by a scalar Young inequality.

Lemma 7.4 (Young form of the fourth-root step)**.** *Let $y, q, M, \eta \ge 0$, with $M > 0$ and $\eta > 0$. If*

$$y^4 \le 3M^4 q,$$

then

$$y \le \eta M + \frac{3}{\eta^3} M q.$$

Proof. Put $t = y/M$. Then $t \ge 0$ and $t^4 \le 3q$. If $t \le \eta$, the claim follows immediately. Otherwise $\eta \le t$, hence

$$t \le \frac{t^4}{\eta^3} \le \frac{3q}{\eta^3}.$$

Multiplying by M and adding the harmless term ηM gives the displayed estimate in both cases. □

Theorem 7.5 (Uniform fourth-power local U^2 criterion for fixed h)**.** *Assume that*

$$\sup_{1\le M\le H} \frac{1}{X} \sum_{X\le x\le 2X-M} \mathcal{U}_2(\tilde{a}_h; I_{x,M}) \to 0 \qquad (X\to\infty). \tag{7.1}$$

Then

$$A_h(N) = o(N) \qquad (N\to\infty).$$

Proof. We first derive the first-moment hypothesis. Fix $\varepsilon > 0$, set $\eta := \varepsilon/2$, and put

$$K := \frac{3}{\eta^3}.$$

Apply (7.1) with the tolerance $\delta := \varepsilon/(2K)$. For all sufficiently large X, uniformly for $1 \le M \le H$,

$$\sum_{X \le x \le 2X-M} \mathcal{U}_2(\widetilde{a}_h; I_{x,M}) \le \delta X.$$

For each such x, M, Proposition 7.2 and Lemma 7.4 give

$$|B_h(x; M)| \le \eta M + KM\mathcal{U}_2(\widetilde{a}_h; I_{x,M}).$$

Summing over $X \le x \le 2X - M$, and using that this interval has at most X integer starting points, gives

$$\sum_{X \le x \le 2X-M} |B_h(x; M)| \le \eta MX + KM \sum_{X \le x \le 2X-M} \mathcal{U}_2(\widetilde{a}_h; I_{x,M}).$$

The last sum is at most δX, hence

$$\sum_{X \le x \le 2X-M} |B_h(x; M)| \le \frac{\varepsilon}{2} MX + KM\frac{\varepsilon}{2K} X = \varepsilon XM.$$

This is precisely (6.2). The conclusion follows from Theorem 6.2. □

Remark 7.6 (Why the natural U^3 upgrade still misses fixed h)**.** A tempting way to force the local input is to rewrite the block sum as a mean of a discrete derivative and then try to control that derivative by a local U^3 norm of λ. On a finite abelian group G one has the exact identity from Gowers-norm formalism [8, 9]

$$\|f\|^8_{U^3(G)} = \mathbb{E}_{r \in G} \|\Delta_r f\|^4_{U^2(G)}, \qquad \Delta_r f(x) := f(x+r)\overline{f(x)}. \tag{7.2}$$

Thus a local U^3 bound for λ controls fourth powers of local U^2 quantities for the derivative sequences $\lambda(n)\lambda(n+r)$ only *after averaging over the shift parameter* r. In schematic form this yields bounds of the shape

$$\frac{1}{X} \sum_x \frac{1}{M} \sum_{|r| \ll M} \|\lambda(\cdot)\lambda(\cdot + r)\|^4_{U^2([x+1,x+M])} = o(1), \tag{7.3}$$

which are consistent with a single exceptional fixed shift $r = h$. Consequently the local U^3 route proves only an averaged-in-r version of the fixed-shift local U^2 criterion, and does not remove the need for a genuinely fixed-shift argument.

8 Bad blocks, projected packets, and the remaining bottleneck

We now turn to the endpoint of the reduction developed above. Before introducing packets, we record the precise reason why the prime-shear symmetry can only hold after projection to the divisible slice.

For a finite interval $J \subset \mathbb{Z}$, define the local Fourier sum

$$A^{(\alpha)}_J(\xi) := \sum_{n \in J} g_\alpha(n)\, e(-\xi n)\,.$$

If $p \nmid h$ is prime, decompose by residue classes modulo p:

$$A_J^{(\alpha)}(\xi) = \sum_{r=0}^{p-1} A_{J,r}^{(\alpha)}(\xi), \qquad A_{J,r}^{(\alpha)}(\xi) := \sum_{\substack{n\in J \\ n\equiv r\ (p)}} g_\alpha(n)\, e(-\xi n)\,.$$

The next proposition is the exact correction to the naive full-block affine-shear statement.

Proposition 8.1 (No direct prime-shear consistency on the full block)**.** *Let $p \nmid h$ be prime and let $J \subset \mathbb{Z}$ be a finite interval. Then the $r = 0$ slice satisfies the exact identity*

$$A_{J,0}^{(\alpha)}(\xi) = -\, A_{J/p}^{(p^2\alpha)}\left(p\xi - \frac{\alpha p(p-1)}{2}\right) + O(1), \tag{8.1}$$

where $J/p := \{m \in \mathbb{Z} :\ pm \in J\}$ and the $O(1)$ term is a boundary error. By contrast, for each $r \neq 0$ one has only the tautological expression

$$A_{J,r}^{(\alpha)}(\xi) = \sum_{\substack{m\in\mathbb{Z} \\ pm+r\in J}} \lambda(pm+r)\, e(Q_\alpha(pm+r) - \xi(pm+r))\,,$$

for which complete multiplicativity gives no simplification. In particular, there is no full-block affine-shear law of the shape

$$A_J^{(\alpha)}(\xi) \approx c_{p,\alpha,\xi}\, A_{J'}^{(\beta)}(T_p\xi)$$

up to boundary error alone, because the contribution of the nonzero residue classes can be as large as the whole sum.

Proof. On the divisible slice, write $n = pm$. Since $\lambda(pm) = -\lambda(m)$, we have

$$A_{J,0}^{(\alpha)}(\xi) = -\sum_{m\in J/p} \lambda(m)\, e(Q_\alpha(pm) - p\xi m) + O(1),$$

where the $O(1)$ term accounts for endpoints when J is not exactly a union of residue classes modulo p. A direct calculation gives

$$Q_\alpha(pm) - Q_{p^2\alpha}(m) = \frac{\alpha p(p-1)}{2}\, m,$$

whence

$$Q_\alpha(pm) - p\xi m = Q_{p^2\alpha}(m) - \left(p\xi - \frac{\alpha p(p-1)}{2}\right) m.$$

Substituting this identity proves (8.1).

For $r \neq 0$, writing $n = pm + r$ yields

$$A_{J,r}^{(\alpha)}(\xi) = \sum_{\substack{m\in\mathbb{Z} \\ pm+r\in J}} \lambda(pm+r)\, e(Q_\alpha(pm+r) - \xi(pm+r))\,,$$

and there is no relation between $\lambda(pm + r)$ and $\lambda(m)$ furnished by complete multiplicativity. Therefore the uncontrolled nonzero residue classes cannot be absorbed into a boundary error, and the full-block sum admits no exact analogue of the divisible-slice shear identity. $\square$

Remark 8.2 (What the preceding proposition means)**.** Proposition 8.1 shows that the arithmetic self-similarity lives only on the sparse slice $n \equiv 0 \pmod p$. For the original derivative sequence $a_h(n) = \lambda(n)\lambda(n + h)$ the situation is even less symmetric, since on that slice one has

$$a_h(pm) = \lambda(pm)\lambda(pm + h),$$

which does not simplify multiplicatively when $p \nmid h$. Thus the correct next target is not a full-block prime-shear law, but a theorem that manufactures a projector or weight selecting enough of the divisible slice while preserving the mass coming from a bad fixed-shift block.

Let $I \subset \mathbb{Z}$ be a finite interval of length L, let $\alpha \in \mathbb{R}$, and define

$$S_I(\alpha) := \sum_{n \in I} a_h(n)\, e(\alpha n)\,.$$

To linearize the shift h, introduce the quadratic phase

$$Q_\alpha(n) := \frac{\alpha}{2h} n(n-h) \qquad (n \in \mathbb{Z}),$$

so that

$$Q_\alpha(n+h) - Q_\alpha(n) = \alpha n.$$

Define

$$g_\alpha(n) := \lambda(n)\, e(Q_\alpha(n))\,.$$

Then

$$g_\alpha(n+h)\overline{g_\alpha(n)} = a_h(n)\, e(\alpha n)\,,$$

and therefore

$$S_I(\alpha) = \sum_{n \in I} g_\alpha(n+h)\overline{g_\alpha(n)}.$$

Proposition 8.3 (A bad block forces an unprojected packet)**.** *Define*

$$\mathcal{D}_{I,\alpha} := \sum_{|r|<L} (L-|r|)\, e(\alpha r) \sum_{n \in \mathbb{Z}} a_h(n+r) a_h(n)\, \mathbf{1}_I(n+r) \mathbf{1}_I(n).$$

Then

$$\mathcal{D}_{I,\alpha} = |S_I(\alpha)|^2.$$

In particular, if $|S_I(\alpha)| \geq \eta L$, then $\mathcal{D}_{I,\alpha} \geq \eta^2 L^2$.

Proof. Expanding the square gives

$$|S_I(\alpha)|^2 = \sum_{u,v \in I} a_h(u) a_h(v)\, e(\alpha(u-v))\,.$$

Write $r = u - v$. For fixed r with $|r| < L$, the number of pairs $(u,v) \in I^2$ with $u - v = r$ is $L - |r|$, and there are no such pairs when $|r| \geq L$. Hence

$$|S_I(\alpha)|^2 = \sum_{|r|<L} (L-|r|)\, e(\alpha r) \sum_{n \in \mathbb{Z}} a_h(n+r) a_h(n)\, \mathbf{1}_I(n+r) \mathbf{1}_I(n),$$

which is exactly $\mathcal{D}_{I,\alpha}$. □

The packet $\mathcal{D}_{I,\alpha}$ is the exact unprojected delta/off-diagonal object forced by a bad block. The next theorem shows that one can already manufacture a genuine projected *delta* packet on a small prime-divisible slice.

Theorem 8.4 (Small-prime selector for a projected delta packet)**.** *Fix $0 < \eta \leq 1$. There exists a function $P_\eta(L) \to \infty$ with $P_\eta(L) = L^{o(1)}$ such that the following holds for all sufficiently large L. Let $I \subset \mathbb{Z}$ be an interval of length L, let $\alpha \in \mathbb{R}$, and assume that*

$$|S_I(\alpha)| \geq \eta L.$$

Then there exists a prime $p \leq P_\eta(L)$ with $p \nmid h$ such that

$$\left| \sum_{\substack{n \in I \\ p \mid n}} a_h(n)\, e(\alpha n) \right| \gg_\eta \frac{L}{p}. \tag{8.2}$$

Consequently the projected delta packet

$$\mathcal{D}_{I,\alpha}^{(p)} := \sum_{r\in\mathbb{Z}}\sum_{n\in\mathbb{Z}} a_h(n+r)a_h(n)\,e(\alpha r)\,\mathbf{1}_I(n+r)\mathbf{1}_I(n)\mathbf{1}_{p|n}\mathbf{1}_{p|n+r} \tag{8.3}$$

has size

$$\mathcal{D}_{I,\alpha}^{(p)} \gg_\eta \frac{L^2}{p^2}. \tag{8.4}$$

Since $p \mid n$ and $p \mid n+r$ force $r = p\ell$, one may also rewrite (8.3) *as*

$$\mathcal{D}_{I,\alpha}^{(p)} = \sum_{\ell\in\mathbb{Z}} e(\alpha p\ell) \sum_{n\in\mathbb{Z}} a_h(n+p\ell)a_h(n)\mathbf{1}_I(n+p\ell)\mathbf{1}_I(n)\mathbf{1}_{p|n}. \tag{8.5}$$

Proof. Choose any function $P_\eta(L) \to \infty$ satisfying

$$\sum_{\substack{p\le P_\eta(L)\\ p\nmid h}} \frac{1}{p} \to \infty \qquad \text{and} \qquad \frac{\pi(P_\eta(L))^2}{L} \to 0 \qquad (L\to\infty),$$

for instance $P_\eta(L) := \exp(\sqrt{\log L})$. Let

$$\mathcal{P} := \{p \le P_\eta(L) :\ p \nmid h\}, \qquad w(n) := \sum_{p\in\mathcal{P}} \mathbf{1}_{p|n}.$$

Write

$$S := \sum_{n\in I} a_h(n)\,e(\alpha n) = S_I(\alpha), \qquad S_p := \sum_{\substack{n\in I\\ p|n}} a_h(n)\,e(\alpha n)\,, \qquad \mu := \frac{1}{L}\sum_{n\in I} w(n).$$

Then

$$\sum_{p\in\mathcal{P}} S_p = \sum_{n\in I} w(n)a_h(n)\,e(\alpha n)\,.$$

Centering w at its mean gives

$$\sum_{p\in\mathcal{P}} S_p = \mu S + \sum_{n\in I}(w(n)-\mu)a_h(n)\,e(\alpha n)\,.$$

Hence, by Cauchy–Schwarz,

$$\left|\sum_{p\in\mathcal{P}} S_p\right| \ge \mu|S| - \left(\sum_{n\in I}(w(n)-\mu)^2\right)^{1/2} L^{1/2} = \mu|S| - \sigma L,$$

where

$$\sigma^2 := \frac{1}{L}\sum_{n\in I}(w(n)-\mu)^2.$$

The standard first and second moment estimates for the divisor-counting weight w give

$$\mu = \sum_{p\in\mathcal{P}} \frac{1}{p} + O\Big(\frac{\pi(P_\eta(L))}{L}\Big)$$

and

$$\sigma^2 \ll \sum_{p\in\mathcal{P}} \frac{1}{p} + \frac{\pi(P_\eta(L))^2}{L}.$$

Because $\sum_{p\in\mathcal{P}} 1/p \to \infty$ and $\pi(P_\eta(L))^2/L \to 0$, we have $\sigma/\mu \to 0$. Therefore, for all sufficiently large L,

$$\left|\sum_{p\in\mathcal{P}} S_p\right| \geq \frac{\eta}{2}\mu L.$$

If every $p \in \mathcal{P}$ satisfied $|S_p| < (\eta/2)N_p(I)$, where

$$N_p(I) := \#\{n \in I : \ p \mid n\},$$

then

$$\sum_{p\in\mathcal{P}} |S_p| < \frac{\eta}{2}\sum_{p\in\mathcal{P}} N_p(I) = \frac{\eta}{2}\mu L,$$

contradicting the previous lower bound and the triangle inequality. Hence there exists $p \in \mathcal{P}$ such that

$$|S_p| \geq \frac{\eta}{2}N_p(I) \gg_\eta \frac{L}{p},$$

which proves (8.2).

Finally,

$$|S_p|^2 = \sum_{u,v\in\mathbb{Z}} a_h(u)a_h(v)\, e(\alpha(u-v))\, \mathbf{1}_I(u)\mathbf{1}_I(v)\mathbf{1}_{p|u}\mathbf{1}_{p|v}.$$

Writing $u = n + r$ and $v = n$ yields exactly (8.3). Therefore

$$\mathcal{D}^{(p)}_{I,\alpha} = |S_p|^2 \gg_\eta \frac{L^2}{p^2},$$

which is (8.4). The representation (8.5) follows because $p \mid n$ and $p \mid n + r$ imply $p \mid r$. $\square$

Remark 8.5 (What the small-prime selector does and does not give)**.** Theorem 8.4 produces a genuine projected delta/off-diagonal object from a bad block. However, its kernel is the bare correlation

$$a_h(n + p\ell)a_h(n)\, e(\alpha p\ell)\,,$$

whereas the exact affine-shear mechanism acts on projected packets built from

$$g_\alpha(n + p\ell h)\overline{g_\alpha(n)}.$$

Thus the theorem does not yet produce the projected Fejér/shear packet required for the quadratic-obstruction argument below. It closes the "manufacture a projector" step only at the delta-packet level, not at the Fejér/shear level.

To bring in the exact arithmetic self-similarity that survives after projection, we now work with the projected Fejér/shear packet on the divisible slice modulo a prime $p \nmid h$.

Proposition 8.6 (Exact affine shear on the divisible slice)**.** *Let $p \nmid h$ be prime, let $R \geq 1$ be an integer, and let $z \in \mathbb{C}$ with $|z| = 1$. Define*

$$\mathcal{F}^{(p)}_{I,\alpha}(R; z) := \sum_{|m|<pR} \left(1 - \frac{|m|}{pR}\right) z^m \sum_{n\in\mathbb{Z}} \mathbf{1}_I(n)\mathbf{1}_I(n + mh)\mathbf{1}_{p|n}\mathbf{1}_{p|n+mh}g_\alpha(n + mh)\overline{g_\alpha(n)}.$$

Also define

$$\beta_p := \frac{\alpha p(p-1)}{2}, \qquad I/p := \{n' \in \mathbb{Z} : \ pn' \in I\}.$$

Then

$$\mathcal{F}^{(p)}_{I,\alpha}(R; z) = \sum_{|\ell|<R} \left(1 - \frac{|\ell|}{R}\right) \left(z^p e(\beta_p h)\right)^\ell \sum_{n'\in\mathbb{Z}} \mathbf{1}_{I/p}(n')\mathbf{1}_{I/p}(n' + \ell h)g_{p^2\alpha}(n' + \ell h)\overline{g_{p^2\alpha}(n')}.$$

Proof. Because $p \nmid h$, the conditions $p \mid n$ and $p \mid n + mh$ imply $p \mid m$. Write $m = p\ell$ and $n = pn'$. By complete multiplicativity of λ,

$$g_\alpha(pm) = -\lambda(m)\, e(Q_\alpha(pm))\,.$$

Also

$$Q_\alpha(pm) = Q_{p^2\alpha}(m) + \beta_p m,$$

so

$$g_\alpha(pm) = -e(\beta_p m)\; g_{p^2\alpha}(m).$$

Substituting $m = p\ell$ and $n = pn'$ into the definition of $\mathcal{F}^{(p)}_{I,\alpha}(R; z)$, the two minus signs cancel and we obtain

$$g_\alpha(p(n' + \ell h))\overline{g_\alpha(pn')} = e(\beta_p \ell h)\; g_{p^2\alpha}(n' + \ell h)\overline{g_{p^2\alpha}(n')}.$$

The stated identity follows. □

8.1 Projected transport defect and rigidity

Fix a prime $p \nmid h$, a dyadic interval $J_0 \subset \mathbb{Z}$ of length $|J_0| = 2^L$, and a phase $\alpha \in \mathbb{T}$. Put

$$\beta_p := \frac{\alpha p(p-1)}{2}, \qquad b(n) := e(-\beta_p h)\; g_\alpha(p(n+h))\overline{g_\alpha(pn)} = g_{p^2\alpha}(n+h)\overline{g_{p^2\alpha}(n)}.$$

Then $|b(n)| = 1$ for every n. For each dyadic subinterval $J \subseteq J_0$, define

$$m_J := \frac{1}{|J|}\sum_{n\in J} b(n), \qquad c_J := \frac{m_J}{|m_J|} \in \mathbb{T} \quad (m_J \neq 0).$$

If J has dyadic children J_-, J_+, define the transport defect

$$\delta(J) := \frac{|m_{J_-}| + |m_{J_+}|}{2} - |m_J| \geq 0.$$

Write $\mathcal{D}^\circ(J_0)$ for the set of proper dyadic subintervals of J_0, and define the total transport defect

$$\mathsf{T}(J_0) := \frac{1}{|J_0|}\sum_{J\in\mathcal{D}^\circ(J_0)} |J|\,\delta(J).$$

Finally, for $m \geq 0$ let

$$J_0(m) := \{n \in J_0 :\; n, n+h, \dots, n+mh \in J_0\}.$$

Proposition 8.7 (Small projected transport defect implies projected rigidity)**.** *Assume that $|m_J| \geq \eta$ for every dyadic subinterval $J \subseteq J_0$, where $0 < \eta \leq 1$. Then with $c := c_{J_0}$ one has*

$$\frac{1}{|J_0|}\sum_{n\in J_0} |b(n) - c| \leq \varepsilon, \qquad \varepsilon := \frac{\sqrt{8L\,\mathsf{T}(J_0)}}{\eta}.$$

Proof. Let $J \subseteq J_0$ be dyadic with children J_-, J_+. Write

$$m_{J_\pm} = a_\pm u_\pm, \qquad a_\pm := |m_{J_\pm}| \in [\eta, 1], \qquad u_\pm \in \mathbb{T},$$

and write $m_J = |m_J| c_J$ with $c_J \in \mathbb{T}$. Since

$$m_J = \frac{m_{J_-} + m_{J_+}}{2},$$

we have

$$4|m_J|^2 = (a_- + a_+)^2 - a_-a_+|u_- - u_+|^2.$$

Also,

$$2\delta(J) = a_- + a_+ - 2|m_J|.$$

Hence

$$2\delta(J) = \frac{a_-a_+|u_- - u_+|^2}{a_- + a_+ + 2|m_J|} \geq \frac{\eta^2}{4}|u_- - u_+|^2,$$

so

$$|u_- - u_+|^2 \leq \frac{8}{\eta^2}\,\delta(J).$$

Because c_J is the phase of the convex combination $a_-u_- + a_+u_+$, it lies on the shorter arc joining u_- and u_+. Therefore

$$|c_{J_\pm} - c_J| \leq |u_- - u_+|, \qquad |c_{J_\pm} - c_J|^2 \leq \frac{8}{\eta^2}\,\delta(J).$$

Now fix $n \in J_0$, and let

$$J_0 \supset J_1(n) \supset \cdots \supset J_L(n) = \{n\}$$

be the dyadic ancestor chain of n. Since $c_{\{n\}} = b(n)$, telescoping gives

$$|b(n) - c_{J_0}| \leq \sum_{j=0}^{L-1} |c_{J_{j+1}(n)} - c_{J_j(n)}|.$$

By Cauchy–Schwarz,

$$|b(n) - c_{J_0}|^2 \leq L\sum_{j=0}^{L-1} |c_{J_{j+1}(n)} - c_{J_j(n)}|^2.$$

Average over $n \in J_0$. Each internal dyadic interval $J \subsetneq J_0$ contributes with weight $|J|/|J_0|$, so

$$\frac{1}{|J_0|}\sum_{n\in J_0} |b(n) - c_{J_0}|^2 \leq \frac{8L}{\eta^2|J_0|}\sum_{J\in\mathcal{D}^\circ(J_0)} |J|\,\delta(J) = \frac{8L}{\eta^2}\,\mathsf{T}(J_0).$$

Taking square roots and using Cauchy–Schwarz once more gives the claim. □

Proposition 8.8 (Projected rigidity yields a long projected Fejér packet)**.** *Assume the hypotheses of Proposition 8.7. Let $R \geq 1$ satisfy*

$$R \leq \frac{|J_0|}{8|h|} \qquad \textit{and} \qquad \mathsf{T}(J_0) \leq \frac{\eta^2}{512LR^2}.$$

Then, with $c := c_{J_0}$,

$$\Re\sum_{m=0}^{R-1}\left(1 - \frac{m}{R}\right)\bar{c}^{\,m}\frac{1}{|J_0(m)|}\sum_{n\in J_0(m)} g_{p^2\alpha}(n+mh)\overline{g_{p^2\alpha}(n)} \geq \frac{R}{4}.$$

Consequently, if $I := pJ_0$ and $z \in \mathbb{T}$ is chosen so that

$$z^p e(\beta_p h) = \bar{c},$$

then

$$\Re\,\mathcal{F}^{(p)}_{I,\alpha}(R; z) \geq \frac{R|J_0|}{8}.$$

Proof. By Proposition 8.7,

$$\frac{1}{|J_0|}\sum_{n\in J_0}|b(n)-c|\le \varepsilon := \frac{\sqrt{8L\,\mathsf{T}(J_0)}}{\eta}\le \frac{1}{8R}.$$

For $0\le m<R$ and $n\in J_0(m)$ one has the telescoping identity

$$g_{p^2\alpha}(n+mh)\overline{g_{p^2\alpha}(n)}\,\overline{c}^m=\prod_{j=0}^{m-1}(b(n+jh)\overline{c}).$$

Since each factor has modulus at most 1,

$$\left|\prod_{j=0}^{m-1}z_j-1\right|\le\sum_{j=0}^{m-1}|z_j-1|\qquad(|z_j|\le 1).$$

Therefore

$$\left|\frac{1}{|J_0(m)|}\sum_{n\in J_0(m)}g_{p^2\alpha}(n+mh)\overline{g_{p^2\alpha}(n)}\,\overline{c}^m-1\right|\le\sum_{j=0}^{m-1}\frac{1}{|J_0(m)|}\sum_{n\in J_0(m)}|b(n+jh)-c|.$$

Because $m<R\le|J_0|/(8|h|)$, we have $|J_0(m)|\ge|J_0|/2$. Hence each inner average is at most 2ε, and so

$$\left|\frac{1}{|J_0(m)|}\sum_{n\in J_0(m)}g_{p^2\alpha}(n+mh)\overline{g_{p^2\alpha}(n)}\,\overline{c}^m-1\right|\le 2m\varepsilon.$$

Taking real parts and summing against Fejér weights yields

$$\Re\sum_{m=0}^{R-1}\left(1-\frac{m}{R}\right)\overline{c}^m\frac{1}{|J_0(m)|}\sum_{n\in J_0(m)}g_{p^2\alpha}(n+mh)\overline{g_{p^2\alpha}(n)}\ge\sum_{m=0}^{R-1}\left(1-\frac{m}{R}\right)(1-2m\varepsilon).$$

Since $R\varepsilon\le 1/8$,

$$\sum_{m=0}^{R-1}\left(1-\frac{m}{R}\right)(1-2m\varepsilon)\ge\frac{R}{2}-2\varepsilon\sum_{m=0}^{R-1}m\ge\frac{R}{2}-\varepsilon R^2\ge\frac{R}{4}.$$

This proves the first claim.

For the second, note that $I=pJ_0$ and Proposition 8.6 give

$$\mathcal{F}^{(p)}_{I,\alpha}(R;z)=\sum_{|m|<R}\left(1-\frac{|m|}{R}\right)(z^pe(\beta_ph))^m\sum_{n\in\mathbb{Z}}\mathbf{1}_{J_0}(n)\mathbf{1}_{J_0}(n+mh)g_{p^2\alpha}(n+mh)\overline{g_{p^2\alpha}(n)}.$$

By the choice of z, the phase factor equals $\overline{c}^m$. Discarding the negative values of m and using $|J_0(m)|\ge|J_0|/2$ for $0\le m<R$, we obtain

$$\Re\,\mathcal{F}^{(p)}_{I,\alpha}(R;z)\ge\frac{|J_0|}{2}\Re\sum_{m=0}^{R-1}\left(1-\frac{m}{R}\right)\overline{c}^m\frac{1}{|J_0(m)|}\sum_{n\in J_0(m)}g_{p^2\alpha}(n+mh)\overline{g_{p^2\alpha}(n)}\ge\frac{R|J_0|}{8}.$$

This proves the proposition. □

Proposition 8.9 (Exact transport-defect identity on a projected slice)**.** *With the notation above,*

$$\sum_{J\in\mathcal{D}^\circ(J_0)}|J|\,\delta(J)=|J_0|-|J_0|\,|m_{J_0}|.$$

Equivalently,

$$\mathsf{T}(J_0)=1-|m_{J_0}|.$$

Proof. By definition,

$$|J|\,\delta(J) = \frac{|J|}{2}\big(|m_{J_-}| + |m_{J_+}|\big) - |J|\,|m_J| = |J_-|\,|m_{J_-}| + |J_+|\,|m_{J_+}| - |J|\,|m_J|.$$

Summing over all proper dyadic subintervals $J \subsetneq J_0$ telescopes to

$$\sum_{J\in\mathcal{D}^\circ(J_0)} |J|\,\delta(J) = \sum_{\text{dyadic leaves } \{n\}\subset J_0} |\{n\}|\,|m_{\{n\}}| - |J_0|\,|m_{J_0}|.$$

But $m_{\{n\}} = b(n)$ and $|b(n)| = 1$, so the leaf sum equals $|J_0|$. This gives the identity. $\square$

Corollary 8.10 (Greedy chain selection from moderate projected bias)**.** *Let $\rho := |m_{J_0}|$, and build a greedy chain*

$$J_0 \supset J_1 \supset J_2 \supset \cdots$$

by always choosing J_{k+1} to be the child of J_k with larger $|m|$. Then for every $r \geq 1$ one has

$$\sum_{k=0}^{r-1} \delta(J_k) \leq 1 - \rho.$$

Consequently, for any threshold $\lambda > 0$,

$$\#\{0 \leq k < r :\ \delta(J_k) > \lambda\} \leq \frac{1-\rho}{\lambda}.$$

In particular, among the first r levels there is a consecutive run of length at least

$$\frac{r}{\lfloor (1-\rho)/\lambda \rfloor + 1} - 1$$

on which every defect is at most λ.

Proof. For each k,

$$|m_{J_{k+1}}| \geq \frac{|m_{(J_k)_-}| + |m_{(J_k)_+}|}{2} = |m_{J_k}| + \delta(J_k).$$

Summing this from $k = 0$ to $r-1$ and using $|m_{J_r}| \leq 1$ gives

$$\sum_{k=0}^{r-1} \delta(J_k) \leq 1 - |m_{J_0}| = 1 - \rho.$$

The bound on the number of indices with $\delta(J_k) > \lambda$ is immediate. If there are at most $q := \lfloor (1-\rho)/\lambda \rfloor$ bad indices among the first r levels, then these indices split the range $\{0, 1, \ldots, r-1\}$ into at most $q+1$ consecutive blocks, so one block has length at least $r/(q+1) - 1$. $\square$

Proposition 8.11 (A large projected packet forces a quadratic obstruction)**.** *Let $J := I/p$. Assume that $R \asymp |J|/|h|$ and that for some $\kappa > 0$ and some unit complex number z one has*

$$\Re\,\mathcal{F}^{(p)}_{I,\alpha}(R; z) \geq \kappa R |J|.$$

Then

$$\sup_{\deg P \leq 2} \left| \frac{1}{|J|} \sum_{n\in J} \lambda(n)\, e(P(n)) \right| \gg_{\kappa,h} 1.$$

Proof. Choose $\gamma \in \mathbb{R}$ such that
$$e(\gamma h) = z^p e(\beta_p h)\,.$$
Define
$$f(n) := g_{p^2\alpha}(n)\mathbf{1}_J(n), \qquad \widetilde{f}(n) := f(n)e(\gamma n)\,.$$
By Proposition 8.6,
$$\mathcal{F}^{(p)}_{I,\alpha}(R;z) = \sum_{|\ell|<R}\Big(1-\frac{|\ell|}{R}\Big)\sum_{n\in\mathbb{Z}}\widetilde{f}(n+\ell h)\overline{\widetilde{f}(n)}.$$
Embed J into a cyclic group $G=\mathbb{Z}/N\mathbb{Z}$ with $N \asymp |J|$ and no wraparound. Let
$$\widehat{\widetilde{f}}(\xi) := \frac{1}{N}\sum_{n\in G}\widetilde{f}(n)\,e(-\xi n/N) \qquad (\xi\in G).$$
Fourier inversion gives
$$\sum_{|\ell|<R}\Big(1-\frac{|\ell|}{R}\Big)\sum_{n\in G}\widetilde{f}(n+\ell h)\overline{\widetilde{f}(n)} = N\sum_{\xi\in G}|\widehat{\widetilde{f}}(\xi)|^2 F_R\Big(\frac{h\xi}{N}\Big)\,,$$
where
$$F_R(t) := \sum_{|\ell|<R}\Big(1-\frac{|\ell|}{R}\Big)e(\ell t)$$
is the Fejér kernel. Hence
$$N\sum_{\xi\in G}|\widehat{\widetilde{f}}(\xi)|^2 F_R\Big(\frac{h\xi}{N}\Big) \ge \kappa R|J|.$$
Since $\sum_{\xi\in G}|\widehat{\widetilde{f}}(\xi)|^2 = |J|/N$ by Parseval and $F_R \ge 0$, there is a constant $c_h > 0$ such that the set
$$\Omega_h := \Big\{\xi\in G : F_R\Big(\frac{h\xi}{N}\Big) \ge c_h R\Big\}$$
has cardinality $O_h(1)$ when $R \asymp N/|h|$. Therefore some $\xi_0 \in \Omega_h$ satisfies
$$|\widehat{\widetilde{f}}(\xi_0)|^2 \gg_{\kappa,h} 1.$$
Unwinding the definitions,
$$\widehat{\widetilde{f}}(\xi_0) = \frac{1}{N}\sum_{n\in J}\lambda(n)\,e\Big(Q_{p^2\alpha}(n)+\gamma n-\xi_0 n/N\Big)\,.$$
The phase
$$P(n) := Q_{p^2\alpha}(n)+\gamma n-\xi_0 n/N$$
has degree at most 2, and $N \asymp |J|$, so
$$\sup_{\deg P\le 2}\left|\frac{1}{|J|}\sum_{n\in J}\lambda(n)\,e(P(n))\right| \gg_{\kappa,h} 1.$$
This proves the claim. $\square$

This conclusion should be viewed as a local quadratic-phase obstruction in the spirit of the polynomial-phase and nilsequence literature [11, 5, 12].

Remark 8.12 (Remaining bottleneck)**.** The rigorous chain established above isolates the remaining obstruction. Indeed,

$$\text{bad block} \Longrightarrow \text{large unprojected packet}$$

by Proposition 8.3, and also

$$\text{bad block} \Longrightarrow \text{large projected delta packet on } 0 \pmod p$$

for some small prime $p \nmid h$, by Theorem 8.4. On the projected side we now also have the deterministic implication

$$\begin{aligned}&\text{tiny projected transport defect} \Longrightarrow \text{projected rigidity}\\ &\Longrightarrow \text{large projected Fejér/shear packet} \Longrightarrow \text{local quadratic obstruction,}\end{aligned}$$

by Propositions 8.7, 8.8, and 8.11.

However, Proposition 8.9 shows that tiny total projected transport defect is equivalent to near-total projected top mean:

$$\mathsf{T}(J_0) = 1 - |m_{J_0}|.$$

Thus a selector that only produces a projected slice with $|m_{J_0}| \gg_\eta 1$ cannot by itself yield the transport bound needed for Proposition 8.8. The strongest unconditional replacement currently available is Corollary 8.10: from moderate projected bias one can still extract a long run of small *one-step* defects along a greedy chain.

The remaining gap is therefore sharper than the earlier "delta packet to Fejér packet" bridge. One must either:

(i) manufacture a projector or weight that boosts the projected top mean from $\gg_\eta 1$ to $1 - o(1)$, so that Proposition 8.8 applies at the full length $R \asymp |J_0|/|h|$; or

(ii) prove a stronger contagion theorem that upgrades a long run of small one-step defects, as in Corollary 8.10, directly into a long projected Fejér packet.

This is the remaining bottleneck in the present argument.

9 Proof of the conditional fixed-shift theorem

Proof of Theorem 1.1. The hypothesis of Theorem 1.1 is precisely the short-interval first-moment assumption (6.2). By Theorem 6.2, this yields (2.2). Theorem 2.1 then applies directly and gives

$$A_h(N) = o(N) \qquad (N \to \infty).$$

□

Remark 9.1. There are two routes through the paper. The direct route proving Theorem 1.1 is

$$\text{absolute first moment} \Longrightarrow \text{critical short-block energy} \Longrightarrow \text{prefix cancellation.}$$

The structural route is

$$\begin{gathered}\text{short-block criterion} \Longrightarrow \text{energy profile and discrete derivative} \Longrightarrow \text{Fourier–Dirichlet}\\ \text{reformulation} \Longrightarrow \text{positive multiscale estimate} \Longrightarrow \text{dyadic local quartic shells} \Longrightarrow\\ \text{fourth-power local } U^2 \text{ criterion} \Longrightarrow \text{packet reformulations and projected bottleneck.}\end{gathered}$$

Theorem 1.1 is conditional.

The paper ends with the projected-packet bottleneck rather than with an unconditional proof of fixed-shift pair Chowla.

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